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A nonparametric measure of independence under a hypothesis of independent components

Murray Rosenblatt

Department of Mathematics, University of California, San Diego, La Jolla, CA, USA

Bruce E. Wahlen

RDT&E Division, Naval Command, Control and Ocean Surveillance Center, San Diego, CA, USA

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Abstract: Asymptotic normality is derived for a nonparametric measure of independence of the components of random two-vectors. This result is obtained without the restrictive assumptions previously made on the rate of convergence of the bandwidth sequence of the density estimates used.

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1. Introduction

Let $X^{(1)}, X^{(2)}, \ldots, X^{(n)}$ be independent and identically distributed random two-vectors $X=(X_1, X_2)$ with joint density function $f(x)=f(x_1, x_2)$ and marginal density functions $g(x_1)$ and $h(x_2)$. We investigate the asymptotic behavior of a nonparametric measure of the independence of X_1 and X_2 based on kernel estimators of f, g, and h which are defined, respectively, as

$$f_n(x) = \frac{1}{nb(n)^2} \sum_{i=1}^{n} W\left(\frac{x - X^{(i)}}{b(n)}\right), \tag{1.1}$$

$$g_n(x_1) = \frac{1}{nb(n)} \sum_{i=1}^{n} w_1\left(\frac{x_1 - X_1^{(i)}}{b(n)}\right), \tag{1.2}$$

Correspondence to: Prof. Murray Rosenblatt, Department of Mathematics, University of California, San Diego, 9500 Gilman Drive, La Jolla, CA 92093-0112, USA.

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and

$$h_n(x_2) = \frac{1}{nb(n)} \sum_{i=1}^{n} w_2\left(\frac{x_2 - X_2^{(i)}}{b(n)}\right). \tag{1.3}$$

Here W is a kernel with the product form $W(x) = w_1(x_1)w_2(x_2)$ and $b = b(n)$ is a sequence of bandwidths such that $b(n) \downarrow 0$ and $nb(n)^2 \to \infty$ as $n \to \infty$.

As a global measure of independence of X_1 and X_2 we consider the quadratic functional

$$I_n = \int [f_n(x) - g_n(x_1)h_n(x_2)]^2 a(x)\, \mathrm{d}x, \tag{1.4}$$

where a is some weight function. Rosenblatt (1975) proved that if the components, X_1 and X_2, are independent then I_n, appropriately centered and normed, is asymptotically normally distributed. He established this fact by using a method of Poissonization of sample size which required the additional assumption that $b(n) = o(n^{-1/5})$ as $n \to \infty$. The purpose of this paper is to prove that under the hypothesis of independent components and under certain conditions on f, w_1, w_2, and a, the functional I_n can be shown to be asymptotically normally distributed without requiring this additional restrictive assumption on the rate of convergence of the bandwidth sequence. For reference we state these conditions now as (A1), (A2) and (A3).

(A1) The probability density function f is supported on all of $\mathbb{R}^2$. Also f and its second order partial derivatives are bounded and uniformly continuous on $\mathbb{R}^2$.

(A2) The kernel functions w_1 and w_2 satisfy $w_1(\cdot) \equiv w_2(\cdot) \equiv w(\cdot)$, where w is a bounded, nonnegative function on $\mathbb{R}$ such that

$$\int w(\alpha)\, \mathrm{d}\alpha = 1, \qquad \int \alpha w(\alpha)\, \mathrm{d}\alpha = 0 \quad \text{and} \quad \int \alpha^2 w(\alpha)\, \mathrm{d}\alpha = 2M < \infty.$$

(A3) The weight function satisfies $a(x) \equiv 1$.

(A2) → w is a probability density function with zero mean and finite variance

To prove asymptotic normality I_n is expanded as the sum of three terms whose asymptotics are derived by various methods. The hypothesis of independent components together with the assumption of a product form for W imply that $Ef_n(x) = Eg_n(x_1)Eh_n(x_2)$, which allows an expansion of I_n as

$$\begin{aligned} I_n &= \int [f_n(x) - Ef_n(x)]^2\, \mathrm{d}x + \int [g_n(x_1)h_n(x_2) - Eg_n(x_1)Eh_n(x_2)]^2\, \mathrm{d}x \\ &\quad - 2\int [f_n(x) - Ef_n(x)][g_n(x_1)h_n(x_2) - Eg_n(x_1)Eh_n(x_2)]\, \mathrm{d}x \\ &= I_{n1} + I_{n2} + I_{n3}. \end{aligned} \tag{1.5}$$

Bickel and Rosenblatt (1973) proved that I_{n1} is asymptotically normal in the univariate case; in the bivariate case we use a central limit theorem of Hall (1984) for centered, degenerate U-statistics which he derived from the martingale results of Brown (1971). Section 2 contains a sequence of lemmas describing the asymptotics of I_{n1}, I_{n2}, and I_{n3}. Section 3 contains proofs of these lemmas and of the main result of this paper which we state below as a theorem. Section 4 contains a brief discussion of the results.

Theorem. *Let $X^{(1)}, X^{(2)}, \ldots, X^{(n)}$ be independent and identically distributed random two-vectors $X = (X_1, X_2)$ with joint density function $f(x_1, x_2)$ and marginal density functions $g(x_1)$ and $h(x_2)$. Let $f_n(x_1, x_2)$, $g_n(x_1)$, and $h_n(x_2)$ be, respectively, kernel estimators of $f(x_1, x_2)$, $g(x_1)$, and $h(x_2)$ as defined*

in (1.1), (1.2), *and* (1.3), *and let* I_n *be the measure of independence of* X_1 *and* X_2 *defined in* (1.4). *Assume that the components of* X *are independent, that* W *has a product form, that assumptions* (A1)–(A3) *hold, and that* $b(n) \downarrow 0$ *and* $nb(n)^2 \to \infty$ *as* $n \to \infty$. *Then* $b^{-1}\{nb^2 I_n - A(n)\}$ *is asymptotically normally distributed with mean zero and variance* $2\nu^2$, *where*

$$A(n) = \left\{\int w^2(x_1)\,\mathrm{d}x_1\right\}^2 - b(n)\int w^2(x_1)\,\mathrm{d}x_1\left\{\int\left[g^2(y_1)+h^2(y_1)\right]\mathrm{d}y_1\right\}$$

and

$$\nu^2 = \int g^2(x_1)\,\mathrm{d}x_1\int h^2(x_2)\,\mathrm{d}x_2$$
$$\times\int\int\left\{\int\int w(u_1)w(u_2)w(u_1+v_1)w(u_2+v_2)\,\mathrm{d}u_1\,\mathrm{d}u_2\right\}^2\mathrm{d}v_1\,\mathrm{d}v_2.$$

2. Asymptotics of I_{n1}, I_{n2}, and I_{n3}

The expression I_n is a sum of functions of quadruples of random variables; while I_{n1} is a sum of functions of pairs of random 2-vectors. Lemma 1, describing the asymptotic normality of I_{n1}, is an almost immediate consequence of Hall's central limit theorem. Lemmas 2 through 5 show that I_{n2} and I_{n3} only make a contribution in mean since their variances are negligible compared to that of I_{n1}.

Lemma 1. I_{n1} *satisfies*

$$I_{n1} = (nb^2)^{-1}\left\{\int w^2(x_1)\,\mathrm{d}x_1\right\}^2 + \mathrm{o_p}\left((nb)^{-1}\right) + 2^{1/2}\nu(nb)^{-1}N_n,$$

where N_n *is asymptotically* N(0, 1) *and* ν^2 *is defined as in the above theorem.*

Lemma 2. *The expectation of* I_{n2} *is given by*

$$E(I_{n2}) = (nb)^{-1}\int w^2(x_1)\,\mathrm{d}x_1\left\{\int\left[g^2(y_1)+h^2(y_1)\right]\mathrm{d}y_1\right\} + \mathrm{o}\left((nb)^{-1}\right).$$

Lemma 3. $E(I_{n3}) = -2E(I_{n2})$.

Lemma 4. $\sigma^2(I_{n2}) = \mathrm{O}((n^2b)^{-1})$.

Lemma 5. $\sigma^2(I_{n3}) = \mathrm{O}((n^2b)^{-1})$.

The variance goes quicly to zero

3. Proofs of results

Some of the details in the following arguments are omitted because of space limitations. The reader is referred to Wahlen (1991, Chapter 2) for these details.

Proof of Lemma 1. This result follows as a special case of a general multivariate result of Hall (1984) in which f is assumed to be a density function on $\mathbb{R}^p$ but the components of the p-vector X are not

$$I_{n2} = \int\left[\frac{1}{n^2}\sum_{i=1}^{n} w\left(\frac{x-x^{(i)}}{b}\right) - \frac{1}{n^2}\sum_{i=1}^{n} E\left(w\left(\frac{x-x^{(i)}}{b}\right)\right)\right]^2 dx = \frac{1}{n^4}\int\left[\sum_{i=1}^{n} w\left(\frac{x-x^{(i)}}{b}\right) - E\left(w\left(\frac{x-x^{(i)}}{b}\right)\right)\right]^2 dx = \frac{1}{n^4}\int\left(\sum_{i=1}^{n} a_i\right)^2 dx$$

$$\left(\sum_{i=1}^{n} a_i\right)^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j = \sum_{i=1}^{n} a_i^2 + \sum_{i\neq j} a_i a_j = \sum_{i=1}^{n} a_i^2 + 2\sum_{i<j}^{n} a_i a_j$$

assumed to be independent. We sketch his argument in the case that $p = 2$. Using the definition of I_{n1} in (1.5) and of $f_n(x)$ in (1.1) we find that

$$\begin{aligned} I_{n1} &= (nb^2)^{-2} \sum_{i=1}^{n} \int \{W_n(x, X^{(i)}) - E \cdots\}^2 \, dx \\ &\quad + 2(nb^2)^{-2} \sum_{1 \leqslant i < j \leqslant n} \int \{W_n(x, X^{(i)}) - E \cdots\}\{W_n(x, X^{(j)}) - E \cdots\} \, dx \\ &= I_{n1}^{(1)} + I_{n1}^{(2)}, \end{aligned} \tag{3.1}$$

where we use $W_n(x, X^{(i)})$ to denote $W((x - X^{(i)})/b(n))$ and an expression of the form $\alpha(X) - E \cdots$ to denote $\alpha(X) - E\alpha(X)$. Now, we observe that $I_{n1}^{(1)} = (nb^2)^{-2} \Sigma_{i=1}^{n} Z_{ni}$, where Z_{ni} is the ith term in the first sum of (3.1). The Z_{ni}'s are independent and identically distributed, and from Lemma 2 of Hall's paper,

$$E(Z_{ni}) = b^2 \int W^2(x) \, dx - b^4 \int \int W(u) W(u+v) \, du \, dv \int f(x) f(x + bv) \, dx$$

and $E(Z_{ni}^2) = O(b^4)$ as $n \to \infty$. Using (3.1) again, we find that $I_{n1}^{(2)} = 2(nb^2)^{-2} U_n$, where U_n is a centered, degenerate U-statistic defined as $U_n = \Sigma_{1 \leqslant i < j \leqslant n} H_n(X^{(i)}, X^{(j)})$ with kernel

$$H_n(X^{(i)}, X^{(j)}) = \int \{W_n(x, X^{(i)}) - E \cdots\}\{W_n(x, X^{(j)}) - E \cdots\} \, dx.$$

From Hall's central limit theorem for such U-statistics it follows that U_n is asymptotically normally distributed with zero mean and variance equal to $\frac{1}{2} n^2 E\{H_n^2(X^{(1)}, X^{(2)})\}$. But, Hall (Lemma 3) showed that $E\{H_n^2(X^{(1)}, X^{(2)})\} \sim b^6 \nu^2$, where ν^2 is defined as in the above theorem. The result now follows from the assumptions that W has a product form, that (A1) and (A2) hold, and that $b(n) \downarrow 0$ and $nb(n)^2 \to \infty$ as $n \to \infty$. □

Proof of Lemma 2. We observe that

$$\begin{aligned} &g_n(x_1) h_n(x_2) - E g_n(x_1) E h_n(x_2) \\ &\quad = [g_n(x_1) - E g_n(x_1)][h_n(x_2) - E h_n(x_2)] \\ &\qquad + [g_n(x_1) - E g_n(x_1)] E h_n(x_2) + [h_n(x_2) - E h_n(x_2)] E g_n(x_1) \\ &\quad = S_1(x) + S_2(x) + S_3(x). \end{aligned} \tag{3.2}$$

Substituting this last expression into the definition of I_{n2} in (1.5) we find that

$$\begin{aligned} I_{n2} &= \int S_1^2(x) \, dx + \int S_2^2(x) \, dx + \int S_3^2(x) \, dx + 2 \int S_1(x) S_2(x) \, dx \\ &\quad + 2 \int S_1(x) S_3(x) \, dx + 2 \int S_2(x) S_3(x) \, dx. \end{aligned} \tag{3.3}$$

We prove the lemma by estimating the expectation of each of the integrals in (3.3).

Results on the variance of density estimates directly imply that $E\{\int S_1^2(x) \, dx\} = o((nb)^{-1})$ and that

$$E\left\{\int S_2^2(x) \, dx\right\} = (nb)^{-1} \int w^2(x_1) \, dx_1 \int h^2(y_1) \, dy_1 + o((nb)^{-1})$$

as $n \to \infty$. A similar result can be derived for $E\{\int S_3^2(x) \, dx\}$ with $g^2(y_1)$ replacing $h^2(y_1)$ in the last equation. Also, the expectation of each of the last three terms on the right-hand side of (3.3) is zero, which proves the lemma. □

Proof of Lemma 3. Substituting (3.2) into the definition of I_{n3} in (1.5) and letting $\Delta_n(x) = f_n(x) - Ef_n(x)$, we find that

$$\begin{aligned} I_{n3} &= -2\int \Delta_n(x) S_1(x)\,\mathrm{d}x - 2\int \Delta_n(x) S_2(x)\,\mathrm{d}x - 2\int \Delta_n(x) S_3(x)\,\mathrm{d}x \\ &= -2I_{n3}^{(1)} - 2I_{n3}^{(2)} - 2I_{n3}^{(3)}. \end{aligned} \tag{3.4}$$

Comparing (3.4) with (3.3) and using the fact that the expectation of each of the last three terms on the right-hand side of (3.3) is zero, we see that it suffices to prove that $E\{I_{n3}^{(1)}\} = \int E\{S_1^2(x)\}\,\mathrm{d}x$, that $E\{I_{n3}^{(2)}\} = \int E\{S_2^2(x)\}\,\mathrm{d}x$, and that $E\{I_{n3}^{(3)}\} = \int E\{S_3^2(x)\}\,\mathrm{d}x$.

Substituting for $\Delta_n(x)$ and $S_1(x)$ into $I_{n3}^{(1)}$, we see that

$$\begin{aligned} E\{I_{n3}^{(1)}\} &= \int E\{[f_n(x) - Ef_n(x)][g_n(x_1) - Eg_n(x_1)][h_n(x_2) - Eh_n(x_2)]\}\,\mathrm{d}x \\ &= (n^2b^4)^{-1} \int E\{[W_n(x, X) - E\cdots][w_n(x_1, X_1) - E\cdots] \\ &\qquad \times [w_n(x_2, X_2) - E\cdots]\}\,\mathrm{d}x, \end{aligned} \tag{3.5}$$

since the two-vectors are independent and identically distributed. But, from the product form of W and the independence of X_1 and X_2, it follows that $EW_n(x, X) = Ew_n(x_1, X_1)Ew_n(x_2, X_2)$ and hence that

$$\begin{aligned} W_n(x, X) - EW_n(x, X) = {} & [w_n(x_1, X_1) - E\cdots][w_n(x_2, X_2) - E\cdots] \\ & + [w_n(x_1, X_1) - E\cdots]Ew_n(x_2, X_2) \\ & + [w_n(x_2, X_2) - E\cdots]Ew_n(x_1, X_1). \end{aligned}$$

Substituting this last expression for $W_n(x, X) - EW_n(x, X)$ into the integrand following the second equality in (3.5) and using the independence of X_1 and X_2, we find that

$$E\{I_{n3}^{(1)}\} = \int \sigma^2\{g_n(x_1)\}\sigma^2\{h_n(x_2)\}\,\mathrm{d}x = \int E\{S_1^2(x)\}\,\mathrm{d}x$$

as was to be proved. It can be shown by similar arguments that $E\{I_{n3}^{(2)}\} = \int E\{S_2^2(x)\}\,\mathrm{d}x$ and that $E\{I_{n3}^{(3)}\} = \int E\{S_3^2(x)\}\,\mathrm{d}x$, which completes the proof of this lemma. □

Proof of Lemma 4. We prove this lemma by estimating the variance of each of the integrals in the expansion of I_{n2} in (3.3). To estimate these variances repeated use is made of the fact that the variance of the product of independent random variables, P and Q, is given by

$$\sigma^2(PQ) = \left[\sigma^2(P) + \{E(P)\}^2\right]\left[\sigma^2(Q) + \{E(Q)\}^2\right] - [E(P)E(Q)]^2. \tag{3.6}$$

Letting $C_n = \int [g_n(x_1) - Eg_n(x_1)]^2\,\mathrm{d}x_1$ and $D_n = \int [h_n(x_2) - Eh_n(x_2)]^2\,\mathrm{d}x_2$ and using (3.6) we find that

$$0 \leqslant \sigma^2\left\{\int S_1^2(x)\,\mathrm{d}x\right\} \leqslant \left[\sigma^2(C_n) + \{E(C_n)\}^2\right]\left[\sigma^2(D_n) + \{E(D_n)\}^2\right]. \tag{3.7}$$

Using Lemmas 2 and 3 of Hall (1984) it can be shown that $E(C_n) = \mathrm{O}((nb)^{-1})$ and $\sigma^2(C_n) = \mathrm{O}((n^2b)^{-1})$, with the same estimates holding for the moments of D_n. It follows from (3.7) that $\sigma^2\{\int S_1^2(x)\,\mathrm{d}x\} = \mathrm{O}((nb)^{-4})$ as $n \to \infty$ and, by similar calculations, that $\sigma^2\{\int S_2^2(x)\,\mathrm{d}x\} = \mathrm{O}((n^2b)^{-1})$ and that $\sigma^2\{\int S_3^2(x)\,\mathrm{d}x\} = \mathrm{O}((n^2b)^{-1})$.

Letting $\tilde{D}_n = \int [h_n(x_2) - Eh_n(x_2)] Eh_n(x_2)\, \mathrm{d}x_2$, and using (3.6) we find that

$$\sigma^2\left\{\int S_1(x) S_2(x)\, \mathrm{d}x\right\} = \left[\sigma^2(C_n) + \{E(C_n)\}^2\right]\sigma^2(\tilde{D}_n), \tag{3.8}$$

since $E(\tilde{D}_n) = 0$. Now, $\tilde{D}_n$ is the mean of independent and identically distributed random variables with zero expectation; therefore,

$$\begin{aligned} 0 \leqslant \sigma^2(\tilde{D}_n) &= (nb^2)^{-1} E\left\{\int Eh_n(x_2)[w_n(x_2, X_2) - E \cdots]\, \mathrm{d}x_2\right\}^2 \\ &= (nb^2)^{-1} \int\int Eh_n(x_2) Eh_n(x_2') \mathrm{Cov}[w_n(x_2, X_2), w_n(x_2', X_2)]\, \mathrm{d}x_2\, \mathrm{d}x_2' \\ &\leqslant (nb^2)^{-1} \int\int Eh_n(x_2) Eh_n(x_2') E\{w_n(x_2, X_2) w_n(x_2', X_2)\}\, \mathrm{d}x_2\, \mathrm{d}x_2', \end{aligned}$$

since $Ew_n(\cdot, X_2)$ and hence $Eh_n(\cdot)$ are each nonnegative by the assumptions of the theorem. Furthermore, Eh_n is uniformly bounded and $\int\int E\{w_n(x_2, X_2) w_n(x_2', X_2)\}\, \mathrm{d}x_2\, \mathrm{d}x_2' = b^2$, from which it follows that $\sigma^2(\tilde{D}_n) = \mathrm{O}(n^{-1})$. Substituting this result and the above moment estimates of C_n into (3.8), it follows that $\sigma^2\{\int S_1(x) S_2(x)\, \mathrm{d}x\} = \mathrm{O}((n^3b^2)^{-1})$ as $n \to \infty$. By a similar argument $\sigma^2\{\int S_1(x) S_3(x)\, \mathrm{d}x\} = \mathrm{O}((n^3b^2)^{-1})$.

Substituting for $S_2(x)$ and $S_3(x)$, letting $\tilde{C}_n = \int [g_n(x_1) - Eg_n(x_1)] Eg_n(x_1)\, \mathrm{d}x_1$, using the above definition of $\tilde{D}_n$ and using (3.6), we find that $\sigma^2\{\int S_2(x) S_3(x)\, \mathrm{d}x\} = \sigma^2(\tilde{C}_n)\sigma^2(\tilde{D}_n)$. But, from the last paragraph we know that $\sigma^2(\tilde{D}_n) = \mathrm{O}(n^{-1})$ and similarly that $\sigma^2(\tilde{C}_n) = \mathrm{O}(n^{-1})$; hence, $\sigma^2\{\int S_2(x) S_3(x)\, \mathrm{d}x\} = \mathrm{O}(n^{-2})$ as $n \to \infty$. Summarizing, we have shown above that the variances of the integrals on the right-hand side of (3.3) are, respectively, $\mathrm{O}((nb)^{-4})$, $\mathrm{O}((n^2b)^{-1})$, $\mathrm{O}((n^2b)^{-1})$, $\mathrm{O}((n^3b^2)^{-1})$, $\mathrm{O}((n^3b^2)^{-1})$, and $\mathrm{O}(n^{-2})$, which proves the lemma. □

Proof of Lemma 5. We prove this lemma by estimating the variance of each of the integrals in the expansion of I_{n3} in (3.4). To carry out the variance calculations each of the three integrals is further expanded as the sum of other terms whose variances we can estimate. These other terms are either: (1) a centered, degenerate U-statistic, or (2) a sum of independent and identically distributed random variables, or (3) a term which can be partitioned into sums of uncorrelated random variables. We note that this proof does not use the assumption of independence of the components of X.

We estimate the variance of $I_{n3}^{(1)}$, the most complicated of the three terms in (3.4), by substituting for $\Delta_n(x)$ and $S_1(x)$ and expressing $I_{n3}^{(1)}$ as

$$\begin{aligned} I_{n3}^{(1)} &= (n^3b^4)^{-1} \sum_{i,j,k=1}^{n} \int [W_n(x, X^{(i)}) - E \cdots][w_n(x_1, X_1^{(j)}) - E \cdots] \\ &\qquad \times [w_n(x_2, X_2^{(k)}) - E \cdots]\, \mathrm{d}x \\ &= (n^3b^4)^{-1}\left\{\sum_{3+0} + \sum_{2+1} + \sum_{1+1+1}\right\}, \end{aligned} \tag{3.9}$$

where Σ_{3+0} denotes the sum when all three indices are equal, Σ_{2+1} denotes the sum when exactly two of the indices are equal, and Σ_{1+1+1} denotes the sum when all three indices are unequal. To estimate the variance of $I_{n3}^{(1)}$ it suffices, therefore, to estimate the variance of each of these three sums.

An argument using the definition of Σ_{3+0} as the sum of independent and identically distributed random variables and the assumptions that w is bounded and nonnegative shows that $\sigma^2\{\Sigma_{3+0}\} = \mathrm{O}(nb^4)$. To estimate the variance of Σ_{1+1+1} we first make an an obvious substitution and partition this sum into

six summands as $\Sigma_{1+1+1} = \Sigma_{i<j<k} a_{ijk} + \Sigma_{i<k<j} a_{ijk} + \cdots + \Sigma_{k<j<i} a_{ijk}$. Since distinct terms of $\Sigma_{i<j<k} a_{ijk}$ are uncorrelated and $E(a_{ijk}) = 0$ if $i<j<k$, then $\sigma^2\{\Sigma_{i<j<k} a_{ijk}\} = \Sigma_{i<j<k} E(a_{ijk}^2)$ and hence

$$\begin{aligned} 0 &\leqslant \sigma^2\Big\{\sum_{i<j<k} a_{ijk}\Big\} \\ &= \frac{n(n-1)(n-2)}{6} \int\int \mathrm{Cov}[W_n(x, X), W_n(x', X)]\mathrm{Cov}[w_n(x_1, X_1), w_n(x_1', X_1)] \\ &\qquad \times \mathrm{Cov}[w_n(x_2, X_2), w_n(x_2', X_2)]\, \mathrm{d}x\, \mathrm{d}x'. \end{aligned}$$

But, $\mathrm{Cov}[w_n(x_1, X_1), w_n(x_1', X_1)] = \mathrm{O}(b)$ by the Cauchy–Schwarz inequality, and thus it follows that $\sigma^2\{\Sigma_{i<j<k} a_{ijk}\} = \mathrm{O}(n^3 b^6)$ and then that $\sigma^2\{\Sigma_{1+1+1}\} = \mathrm{O}(n^3 b^6)$.

To complete the estimate of the variance of $I_{n3}^{(1)}$ it remains only to estimate the variance of Σ_{2+1} which we accomplish by first expressing the sum as

$$\begin{aligned} \sum_{2+1} &= \sum_{i\neq j} \int \big[W_n(x, X^{(i)}) - E\cdots\big]\big[w_n(x_1, X_1^{(i)}) - E\cdots\big]\big[w_n(x_2, X_2^{(j)}) - E\cdots\big]\, \mathrm{d}x \\ &\quad + \sum_{i\neq j} \int \big[W_n(x, X^{(i)}) - E\cdots\big]\big[w_n(x_2, X_2^{(i)}) - E\cdots\big]\big[w_n(x_1, X_1^{(j)}) - E\cdots\big]\, \mathrm{d}x \\ &\quad + \sum_{i\neq j} \int \big[W_n(x, X^{(i)}) - E\cdots\big]\big[w_n(x_1, X_1^{(j)}) - E\cdots\big]\big[w_n(x_2, X_2^{(j)}) - E\cdots\big]\, \mathrm{d}x \\ &= \sum_{2+1} (1) + \sum_{2+1} (2) + \sum_{2+1} (3). \end{aligned} \tag{3.10}$$

We then expand each of these three sums and estimate the variance of each of the terms in the expansions.

By a simple algebraic identity we see that

$$\begin{aligned} \sum_{2+1} (1) &= \sum_{i\neq j} \int \big[W_n(x, X^{(i)}) w_n(x_1, X_1^{(i)}) - E\cdots\big]\big[w_n(x_2, X_2^{(j)}) - E\cdots\big]\, \mathrm{d}x \\ &\quad - \sum_{i\neq j} \int \big[W_n(x, X^{(i)}) - E\cdots\big]\big[w_n(x_2, X_2^{(j)}) - E\cdots\big] E w_n(x_1, X_1)\, \mathrm{d}x \\ &\quad - \sum_{i\neq j} \int \big[w_n(x_1, X_1^{(i)}) - E\cdots\big]\big[w_n(x_2, X_2^{(j)}) - E\cdots\big] E W_n(x, X)\, \mathrm{d}x \\ &\quad + \sum_{i\neq j} \int \big[w_n(x_2, X_2^{(j)}) - E\cdots\big]\mathrm{Cov}[W_n(x, X), w_n(x_1, X_1)]\, \mathrm{d}x \\ &= \sum_{2+1} (1\mathrm{A}) - \sum_{2+1} (1\mathrm{B}) - \sum_{2+1} (1\mathrm{C}) + \sum_{2+1} (1\mathrm{D}). \end{aligned}$$

Now, $\Sigma_{2+1}(1\mathrm{A})$, $\Sigma_{2+1}(1\mathrm{B})$, and $\Sigma_{2+1}(1\mathrm{C})$ can each be partitioned into two sums of uncorrelated random variables. After such partitioning, it can be shown that $\sigma^2\{\Sigma_{2+1}(1\mathrm{A})\} = \mathrm{O}(n^2 b^5)$, that $\sigma^2\{\Sigma_{2+1}(1\mathrm{B})\} = \mathrm{O}(n^2 b^7)$, and that $\sigma^2\{\Sigma_{2+1}(1\mathrm{C})\} = \mathrm{O}(n^2 b^6)$. The last sum above, $\Sigma_{2+1}(1\mathrm{D})$, is the sum of independent and identically distributed random variables and hence $\sigma^2\{\Sigma_{2+1}(1\mathrm{D})\} = \mathrm{O}(n^3 b^5)$. It follows from these estimated variances that $\sigma^2\{\Sigma_{2+1}(1)\} = \mathrm{O}(n^3 b^5)$, and by a similar analysis that $\sigma^2\{\Sigma_{2+1}(2)\} = \mathrm{O}(n^3 b^5)$ also.

Finally, we expand $\Sigma_{2+1}(3)$ as the sum of four terms using a substitution similar to that employed in the expansion of $\Sigma_{2+1}(1)$ and estimate the variance of each of these terms. One of the terms in the expansion of $\Sigma_{2+1}(3)$ equals twice a centered, degenerate U-statistic with variance of order $O(n^2b^6)$; two other terms in the expansion can each be partitioned into two sums of uncorrelated random variables and shown to have variance of order $O(n^2b^7)$; and the fourth term in the expansion is the sum of independent and identically distributed random variables and has variance of order $O(n^3b^8)$. It follows then that $\sigma^2\{\Sigma_{2+1}(3)\} = O(n^3b^8)$ and from (3.10) that $\sigma^2\{\Sigma_{2+1}\} = O(n^3b^5)$. Using (3.9) and the above estimates of the variance of Σ_{3+0}, Σ_{2+1}, and Σ_{1+1+1}, we find that $\sigma^2\{I_{n3}^{(1)}\} = O((nb)^{-3})$.

After expanding $I_{n3}^{(2)}$ in a manner similar to the expansion of $I_{n3}^{(1)}$ in (3.9), it can be shown that $\sigma^2\{I_{n3}^{(2)}\} = O((n^2b)^{-1})$ and likewise that $\sigma^2\{I_{n3}^{(3)}\} = O((n^2b)^{-1})$. But, we just proved that $\sigma^2\{I_{n3}^{(1)}\} = O((nb)^{-3})$. It follows therefore from (3.4) that $\sigma^2(I_{n3}) = O((n^2b)^{-1})$. □

Proof of Theorem. The result follows by a straightforward argument based on the expansion of I_n in (1.5) and Lemmas 1–5. □

4. Discussion

We note that after correction of the centering constant in Rosenblatt (1975, Theorem 2), those results are consistent with the results in this paper. Also, it is noteworthy from a testing viewpoint that asymptotic normality of I_n still holds when using estimators of g and h in the expressions for $A(n)$ and ν^2 in our theorem. Finally, in some applications it would be more natural to assume that the density function f has bounded support rather than assuming, as in (A1), that it is supported on all of $\mathbb{R}^2$. The assumption of bounded support, however, gives rise to a 'boundary effect' problem. We believe this problem can be dealt with in the present context of a product kernel by generalizing the modifications for 'boundary effect' in one dimension, such as the use of a linear combination of kernels of different bandwidths (Rice, 1984), the use of modified boundary kernels (Müller, 1988), or the use of kernels which 'automatically' deform near the boundary (Messer and Goldstein, 1989).

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